



Project Report

Smart Bus Tracking, Speed Monitoring and Digital Ticketing System in Bangladesh

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Table of Contents:

Table of Contents:	2
1. Project Charter:	3
2.1 Identification of Stakeholders	7
2. Stakeholder Analysis Table	9
2.1 Stakeholder Risks	11
2.2 Stakeholder Communication Plan	11
3. Risk Analysis:	11
1. Risk Identification	11
2. Risk Analysis Table	12
3. Risk Matrix (Probability vs Impact)	15
4. Risk Mitigation Strategy (Detailed)	15
4. Cost Benefit Analysis :	17
Cost & Benefit Summary	18
Net Present Value (NPV)	18
Interpretation	18
Internal Rate of Return (IRR)	18
Interpretation	18
Benefit-Cost Ratio (BCR)	19
Interpretation	19
5. Project Scheduling and Planning:	20
5.1 Project Gantt Chart Preparation	20
5.2 Network Diagram	21
5.3 Critical Path Identification	22
5.4 Resource Allocation	23
5.4.1. Resource sheet :	23
5.4.2 Manpower Allocation :	23
5.5 Resource Overallocation and Leveling	25
5.5.1 Resource Overallocation	25
5.5.2 Resource Leveling	26
6. Swot Analysis:	27
Overview:	27
Conclusion:	32
Member is responsible for:	33

1. Project Charter:

PROJECT CHARTER	
Project Name: Smart Bus Tracking, Speed Monitoring and Digital Ticketing System in Bangladesh	
Project Owner: Bus Transport Company Authority	Decision Making Exec: Bangladesh Road Transport Authority (BRTA)
Project Sponsor: Public-Private Partnership (BRTA + Bus Companies)	Project Manager: Sadman Tayef
Customer: Passengers (daily commuters, students, office workers), Bus Transport Companies (Green Line, Shohagh, Hanif), and Public Transport Operators (BRTA, BRTC)	Charter Date: 24 th January 2026
Expected Start Date: 27 April 2026	Expected Completion Date: 7 November 2029
Project Description: This project aims to develop a smart transportation system in Bangladesh that provides real-time bus tracking, speed monitoring, and digital ticketing services. The system will use GPS devices installed in buses to track location and speed, while a mobile application will allow passengers to monitor buses, view speed and estimated arrival time, and purchase tickets digitally using bKash and Nagad.	
Project Mission: To develop a real-time bus tracking, speed monitoring, and digital ticketing solution that improves passenger safety, reduces waiting time, and enhances public transport efficiency in Bangladesh.	

Project Objectives (Specific & Measurable):

- Develop mobile application within 12 months
- Achieve 90% tracking accuracy
- Display real-time speed with ± 5 km/h accuracy
- Reduce waiting time by 30%
- Detect over speeding
- Integrate digital payments
- Onboard at least 50 buses per transport company (pilot phase)

Project Scope and Schedule:

KEY Activities and Milestones	Prerequisite	Duration of the Activity
START	-	0 days
Market Analysis	1	60 days
Project Initiation	2	30 days
Stakeholder & Requirement Analysis	3	60 days
Feasibility Study & Budget Approval	3	60 days
Vendor Selection for GPS Devices	4FS,5FS	60 days
System Architecture Design	4FS-10 days	90 days
UI/UX Design (Mobile App)	4FS-5 days	60 days
Database & Server Setup	7FS-7 days	90 days
Mobile App Development	7FS,8FS-10 days	150 days

Admin Dashboard Development	7FS,9FS	120 days
Payment Gateway Integration	10FS-5 days	60 days
GPS Device Procurement	6FS	60 days
GPS Installation (Pilot Buses)	11FS,13FS	90 days
Tracking & Speed Monitoring Integration	9FS,10FS,13FS	90 days
System Testing	12FS,14FS,15FS	90 days
User Training (Drivers & Admin)	14FS-5 days	60 days
Pilot Deployment (50 buses/company)	16FS+15 days,17FS	60 days
Documentation	16FS	60 days
Final Review & Closing	18FS,19FS	30 days
Operation & Maintenance	20FS	120 days
END	21	0 days

Project Scope and Schedule

COST TYPE	VENDOR / LABOR NAMES	RATE	QTY	Total	Year1	Year2
LABOR	Project Manager, System Analyst, Software Developer, UI/UX Designer, Tester, Network Engineer	Mixed monthly salary	14 staff	₹30,00,000	₹18,00,000	₹12,00,000
Materials	GPS Devices, SIM/Data modules, installation accessories	₹12,000/ bus approx.	50 buses/ Company pilot	₹6,00,000	₹2,00,000	₹4,00,000
Expert Services	Payment Gateway Integration, Security Audit, Cloud/Server Specialist	Fixed service cost	3 services	₹8,00,000	₹3,00,000	₹5,00,000
Consultants	Transport Consultant, Legal/Regulatory Consultant, Business Analyst	Fixed consultan cy cost	3 consultant s	₹5,00,000	₹4,00,000	₹1,00,000
ETC	Server & Hosting, Software Tools, Testing Tools, App Deployment	Fixed technical cost	1 package	₹7,00,000	₹3,00,000	₹4,00,000
MISCELLANEOUS	Training, Awareness Campaign, Documentation, Travel, Contingency	Fixed support cost	1 package	₹4,00,000	₹2,00,000	₹2,00,000
		TOTAL COSTS		₹60,00,000	₹32,00,000	₹28,00,000

2.1 Identification of Stakeholders

Project Stakeholders:

Primary Stakeholders(Directly affected):

- Passengers (daily commuters, students, office workers)
- Bus Transport Companies (Green Line, Shohagh, Hanif)
- Public Transport Authorities (BRTA, BRTC)

Secondary Stakeholders (Indirectly affected):

- Drivers and Bus Staff
- Payment Service Providers (bKash, Nagad)
- Mobile Network Providers
- IT & Development Team
- Government & Regulatory Bodies

Critical Success Factors, Constraints & Assumptions :

Critical Success Factors (Enablers)

- Accurate GPS tracking system
- Reliable mobile application performance
- Strong and stable internet connectivity
- Secure and fast digital payment system
- User-friendly interface for passengers and drivers
- Proper training for drivers and operators

Constraints

- Limited project budget (₹60,00,000)
- Infrastructure limitations in rural areas
- Resistance from traditional transport operators
- Dependency on internet connectivity
- Time constraint for implementation

Assumptions

- Users have smartphones and internet access
- Government will support digital transport initiatives
- Payment systems (bKash, Nagad) will be available
- Transport companies will cooperate during pilot phase
- GPS devices will function properly

Other Related Projects / Initiatives

Digital Bangladesh Initiative

- Existing ride-sharing platforms (Uber, Pathao)
- GPS-based tracking systems (GomaxTracker)
- Smart City initiatives in Bangladesh

Project Owner:

Project Manager:

Signature & Date:

Signature & Date:

2. Stakeholder Analysis Table

Stakeholders	Power/Influence (1 thru 5)	Current Support/ Favorability (1 thru 5)	Calculated Rating	What do they need? What do they consider to be “Wins”? (Success Criteria)	How will we engage them to increase their favorability / support?	Action	Lead	Due Date
Passengers	3	2	6	Real-time tracking, accurate ETA, safe travel, easy digital ticketing	App testing, feedback surveys, notifications, customer support	Improve usability and gather continuous feedback	Product/UI Team	Pilot phase & ongoing
Bus Transport Companies	5	2	10	Higher efficiency, better fleet monitoring, increased revenue, satisfied passengers	Weekly meetings, pilot demos, ROI reports, decision involvement	Align business goals and onboard pilot buses	Project Manager	Before pilot deployment
BRTA / BRTC	5	3	15	Regulatory compliance, safety monitoring, transparent data reporting	Formal reports, approval meetings, compliance documentation	Obtain approvals and maintain policy alignment	Compliance Lead	Before system approval
Government & Regulatory Bodies	5	3	15	Policy support, legal approval, public safety, compliance with national transport rules	Official updates, progress reports, compliance meetings, approval follow-ups	Keep satisfied and secure required approvals	Policy/Compliance Lead	Before approval & periodic
Drivers & Bus Staff	2	3	6	Easy operation, minimal complexity, proper training, quick issue resolution	Hands-on training, support sessions, user guidance	Conduct training and collect operational issues	Training Coordinator	Before pilot deployment
Payment Service Providers (bKash, Nagad)	3	2	6	Secure integration, reliable transactions, fast payment processing	Technical workshops, API testing, service coordination	Complete payment integration and testing	Integration Lead	Before launch
IT & Development Team	4	1	4	Clear requirements, timeline clarity, technical resources, project success	Daily stand-ups, sprint planning, requirement reviews	Finalize requirements and manage development execution	Technical Lead	Ongoing
Mobile Network Providers	2	3	6	Stable network performance, service usage growth, connectivity reliability	Coordination meetings, escalation channels, route coverage reviews	Ensure stable connectivity on pilot routes	Infrastructure Lead	Before pilot routes go live

Power/Influence	Current Support / Favorability
1 = No power / influence over project outcomes	1 = Active supporter
2 = Some influence over project outcomes	2 = Moderately positive, passive support
3 = Moderate influence over project outcomes	3 = Neutral
4 = Major influence over project outcomes	4 = Moderately negative, passive resistance
5 = Maximum influence / mandatory support	5 = Negative, actively working against
Power / Influence Range	Current Support Favorability Range
1	1
2	2
3	3
4	4
5	5

Stakeholder Rating	Thresholds
Major (must take action)	15 - 25
Minor (should take action)	6 - 12
Insignificant (observe)	1 - 5

2.1 Stakeholder Risks

Stakeholder	Risk	Mitigation
Passengers	Low adoption	Awareness campaign
Bus Companies	Resistance	Demonstrate benefits
Regulators	Approval delay	Early engagement
Drivers	Misuse	Training
Payment Providers	Transaction failure	Backup system

2.2 Stakeholder Communication Plan

Stakeholder	Communication Method	Frequency
Bus Companies	Meetings, reports	Weekly
BRTA	Official reports	Monthly
Passengers	App notifications	Daily
IT Team	Internal meetings	Daily
Drivers	Training sessions	Initial + periodic

3. Risk Analysis

This risk analysis evaluates the major threats that may affect the successful implementation of the Smart Bus Tracking, Speed Monitoring and Digital Ticketing System in Bangladesh. The analysis covers technical, operational, financial, market, security, and external risks, and proposes suitable mitigation, monitoring, and ownership plans.

1. Risk Identification

Technical Risks

- GPS failure or inaccurate tracking
- Speed data inaccuracy
- App crashes or bugs
- Server downtime

Operational Risks

- Drivers not using the system properly
- GPS device damage
- Poor internet connectivity

Financial Risks

- Budget overrun
- High maintenance cost
- Low return on investment (ROI)

Market and User Risks

- Low passenger adoption
- Resistance from transport companies
- Lack of digital awareness

Security Risks

- Data breach
- Payment fraud
- Unauthorized system access

External Risks

- Government policy changes
- Traffic conditions
- Power outages

2. Risk Analysis Table

Risk Event Description and Impact	Probability	Severity	Risk Factor	Mitigation Strategy	Who/When
GPS failure - GPS failure can result in inaccurate or missing vehicle tracking data	Medium	High	Poor satellite signal, hardware faults, urban obstruction, device damage	Backup tracking system	IT Team / Maintenance Team - During operation, monitored continuously
Speed inaccuracy - Speed inaccuracy may cause incorrect monitoring and false violation detection	Medium	Medium	Sensor errors, GPS lag, calibration issues, poor signal	Calibration and testing	Technical Team / Q/A Team - During testing and live monitoring

App crash - App crashes can disrupt user access and reduce system usability	Medium	High	Bugs, poor coding, overload, device incompatibility	Regular updates and testing	Development Team - During testing and continuously after deployment
Server downtime- Server downtime may result in system unavailability and service interruption	Low	Medium	High traffic, hardware failure, poor maintenance, cyber attacks	Cloud backup server	IT/Server Admin - 24/7 monitoring during operation
Driver misuse -Driver misuse can lead to safety risks and unreliable system data	Medium	Medium	Lack of training, low accountability, weak enforcement	Training programs	Operations Manager / Supervisors - During daily operations
Internet issue - Internet issues may cause delays in real-time data transmission and system disruption	High	High	Weak network coverage, ISP failure, bandwidth limitations	Offline data buffering	IT Team / Network Provider - Continuous monitoring
Budget overrun - Budget overrun can result in financial pressure and potential project delays	Medium	High	Poor estimation, unexpected costs, scope changes	Cost monitoring	Project Manager / Finance Team - During planning and throughout project
Low adoption - Low adoption may reduce system effectiveness and overall project success	Medium	High	Lack of awareness, poor usability, resistance to change	Awareness campaign	Marketing Team / Project Manager - During launch and early stages
Company resistance - Company resistance can	Medium	High	Fear of change, cost concerns, lack of trust	Demonstrate benefits	Stakeholder Manager / Project Lead -

slow down implementation and limit system adoption					Before and during implementation
Data breach - Data breach may result in loss of sensitive information and legal consequences	Low	Medium	Weak security, poor encryption, phishing attacks	Encryption and firewall	Cybersecurity Team / IT Team - Continuous monitoring
Payment failure - Payment failure can cause transaction errors and revenue loss	Low	Medium	Gateway issues, server errors, network problems	Reliable gateway (bKash/Nagad)	Finance Team / IT Team - During transactions, real-time monitoring
Policy change - Policy changes may require project adjustments and create compliance challenges	Low	Low	New laws, transport regulations, political changes	Flexible system design	Legal Team / Project Manager - Periodically and when updates occur
Power outage - Power outage can lead to system downtime and operational interruption	Medium	Medium	Grid failure, lack of backup power, infrastructure issues	Backup power	IT Team / Facility Management - During operation, ensure backup readiness

3. Risk Matrix (Probability vs Impact)

The following visual matrix places the identified risks according to their probability and impact. High-risk items require immediate management attention.

Probability / Impact	Low	Medium	High
High	-	Internet issue	-
Medium	-	Speed inaccuracy Driver misuse Power outage	GPS failure App crash Budget overrun Low adoption Company resistance
Low	-	Policy change	Server downtime Data breach Payment failure

Legend: Green = Low risk, Yellow = Medium risk, Red = High risk

4. Risk Mitigation Strategy (Detailed)

Technical Mitigation

- Use high-quality GPS devices
- Regular system testing
- Maintain backup server

Operational Mitigation

- Conduct driver training sessions
- Provide clear operating instructions
- Ensure technical support

Financial Mitigation

- Track budget regularly
- Control unnecessary expenses
- Use phased implementation

Market Mitigation

- Run awareness campaigns
- Give demos to bus companies
- Offer incentives for early users

Security Mitigation

- Apply data encryption
- Use secure payment gateway
- Enable user authentication

Activity	Frequency
System performance check	Daily
GPS accuracy check	Weekly
Financial review	Monthly
Security audit	Quarterly

6. Risk Ownership

Risk Category	Responsible
Technical risks	IT Team
Operational risks	Project Manager
Financial risks	Project Owner
Security risks	Security Team

4. Cost Benefit Analysis :

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Task Name	Start	Finish	Duration	DF(12%)	Cost	PVC1	Income	PV11	DF (18%)	PVC2	PVI2			
2	START	Mon 4/27/26	Mon 4/27/26	0	1	0	0	0	0	1	0	0			
3	Market Analysis	Mon 4/27/26	Fri 7/17/26	0.16438	0.98154	360,000.00	353356		0	0.97316	350337	0			
4	Project Initiation	Mon 7/20/26	Fri 8/28/26	0.24658	0.97244	105,000.00	102106		0	0.96001	100801	0			
5	Stakeholder & Requirement Analysis	Mon 8/31/26	Fri 11/20/26	0.41096	0.95449	150,000.00	143174		0	0.93424	140136	0			
6	Feasibility Study & Budget Approval	Mon 8/31/26	Fri 12/4/26	0.60274	0.93397	360,000.00	336230		0	0.90505	325819	0			
7	Vendor Selection for GPS	Mon 12/7/26	Fri 2/26/27	0.76712	0.91673	360,000.00	330025		0	0.88076	317074	0			
8	System Architecture Design	Mon 11/9/26	Fri 3/12/27	1.0137	0.89147	427,500.00	381104		0	0.84554	361468	0			
9	UI/UX Design (Mobile App)	Mon 11/9/26	Fri 1/29/27	1.17808	0.87502	120,000.00	105002		0	0.82284	98741.2	0			
10	Database & Server Setup	Thu 3/4/27	Wed 7/7/27	1.42466	0.85091	202,500.00	172308		0	0.78994	159962	0			
11	Mobile App Development	Mon 3/15/27	Fri 10/8/27	1.83562	0.81218	675,000.00	548224		0	0.73799	498145	0			
12	Admin Dashboard Development	Thu 7/8/27	Wed 12/22/27	2.16438	0.78248	570,000.00	446014		0	0.69891	398377	0			
13	Payment Gateway Integration	Mon	Fri 12/31/27	2.32877	0.76804	150,000.00	115206		0	0.68015	102022	0			
14	GPS Device Procurement	Mon 3/1/27	Fri 5/21/27	2.49315	0.75386	270,000.00	203543		0	0.66189	178711	0			
15	GPS Installation (Pilot Buses)	Thu 12/23/27	Wed 4/26/28	2.73973	0.73309	315,000.00	230923		0	0.63542	200158	0			
16	Tracking & Speed Monitoring Integration	Mon 10/11/27	Fri 2/11/28	2.9863	0.71289	427,500.00	304759		0	0.61001	260780	0			
17	System Testing	Thu 4/27/28	Wed 8/30/28	3.23288	0.69324	135,000.00	93587.5		0	0.58562	79058.4	0			
18	User Training (Drivers & Admin)	Thu 4/20/28	Wed 7/12/28	3.39726	0.68045	165,000.00	112274		0	0.5699	94033.4	0			
19	Pilot Deployment (50 buses/company)	Thu 9/21/28	Wed 12/13/28	3.56164	0.66789	360,000.00	240439		0	0.5546	199657	0			
20	Documentation	Thu 8/31/28	Wed 10/11/28	3.64384	0.66169	660	436.718		0	0.54711	361.092	0			
21	Final Review and Closing	Thu 12/14/28	Wed 1/24/29	3.72603	0.65556	180,000.00	118001		0	0.53972	97149	0			
22	Operation & Maintenance	Thu 1/25/29	Fri 5/25/29	4.05479	0.63158	0	0		0	0.51113	0	0			
23	END	Fri 5/25/29	Fri 5/25/29	0	1	0	0		0	1	0	0			
24	total revenue in 10 years (assumption)	Fri 5/25/29	Sun 5/25/39	14.0548	0.20335			95016840	19321970	0.09766		9279269			
25															
26															
27															
28	Discount rate, r1 = 12%														
29	Discount rate, r2 = 18%														
30	Summation of PVC1							55,333,160							
31	Summation of PV11							95,016,840							
32	Summation of PVC2							13,632,679							
33	Summation of PVI2							9,279,269							
34	XNPV1 (Net Present Value)							89,683,680							
35	XNPV2 (Net Present Value)							84,353,409							
36	XBCR (Benefit-Cost Ratio)							17.82x							
37	XIRR (Internal Rate of Return)							17.72%							
38															

This presents a project cost - benefit analysis broken down into multiple tasks, including phases such as market analysis, project initiation, stakeholder analysis, and feasibility study. It evaluates costs and benefits using discounted cash flow methods at 12% and 18% discount rates.

The project spans **April 2026 - November 2029**, with **benefits projected over 10 years**.

Cost & Benefit Summary

- **Total Present Cost (PVC) = ₪5,333,160**
- **Total Present Benefit (PVI) = ₪95,016,840 (undiscounted)**
- Discounted benefits:
 - **PVI (12%) = ₪95,016,840**
 - **PVI (18%) = ₪9,279,269**

Net Present Value (NPV)

At **12% discount rate**:

$$\text{NPV} = 95,016,840 - 5,333,160 = 89,683,680$$

At **18% discount rate**:

$$\text{NPV} = 9,279,269 - 13,632,679 = -4,353,409$$

Interpretation

- **Positive NPV at 12% → Highly profitable**
- **Negative NPV at 18% → Not viable under high cost of capital**

Internal Rate of Return (IRR)

From sheet:

- **IRR $\approx$ 17.72%**

Interpretation

- Project is acceptable if required return $< 17.72\%$
- Borderline if financing cost approaches this level

Benefit-Cost Ratio (BCR)

- $BCR \approx 17.82$

Extremely high → benefits far exceed costs

Interpretation

- Extremely fast recovery (almost immediate)
- Very low liquidity risk

Overall Financial Insight:

- Strong profitability under moderate discount rate (12%)
- High IRR confirms investment attractiveness
- Benefits are heavily **long-term**, making results sensitive to discount rate
- At **18%**, project becomes **financially unviable**

5. Project Scheduling and Planning

5.1 Project Gantt Chart Preparation

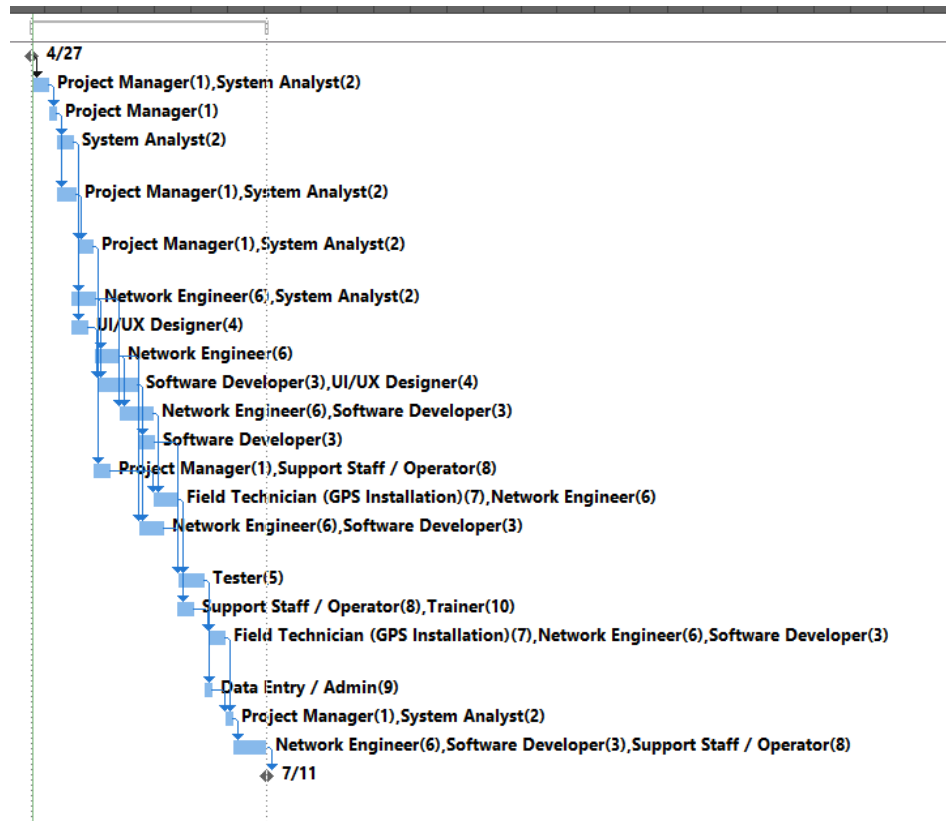


Fig 1: visual representation of the project schedule

The total project duration is approximately **438 days**, mainly due to the long Operation & Maintenance phase. Excluding this phase, the core system development and deployment are completed within a significantly shorter timeframe.

The schedule includes both sequential and parallel activities. For example, *Stakeholder Analysis* and *Feasibility Study* are performed in parallel to reduce delays. Similarly, *UI/UX Design* and *System Architecture Design* overlap to accelerate development.

Lead and lag times (e.g., FS-10 days, FS-5 days) are used to optimize scheduling and reduce idle time between dependent tasks.

In this project, the Gantt Chart was developed using **Microsoft Project** to ensure proper scheduling and tracking.

The Gantt Chart helps in monitoring project progress and ensures timely completion by identifying task dependencies and overlaps.

- Each task is connected logically using **Finish-to-Start (FS)** relationships
- Parallel tasks are identified to reduce total project duration

5.2 Network Diagram

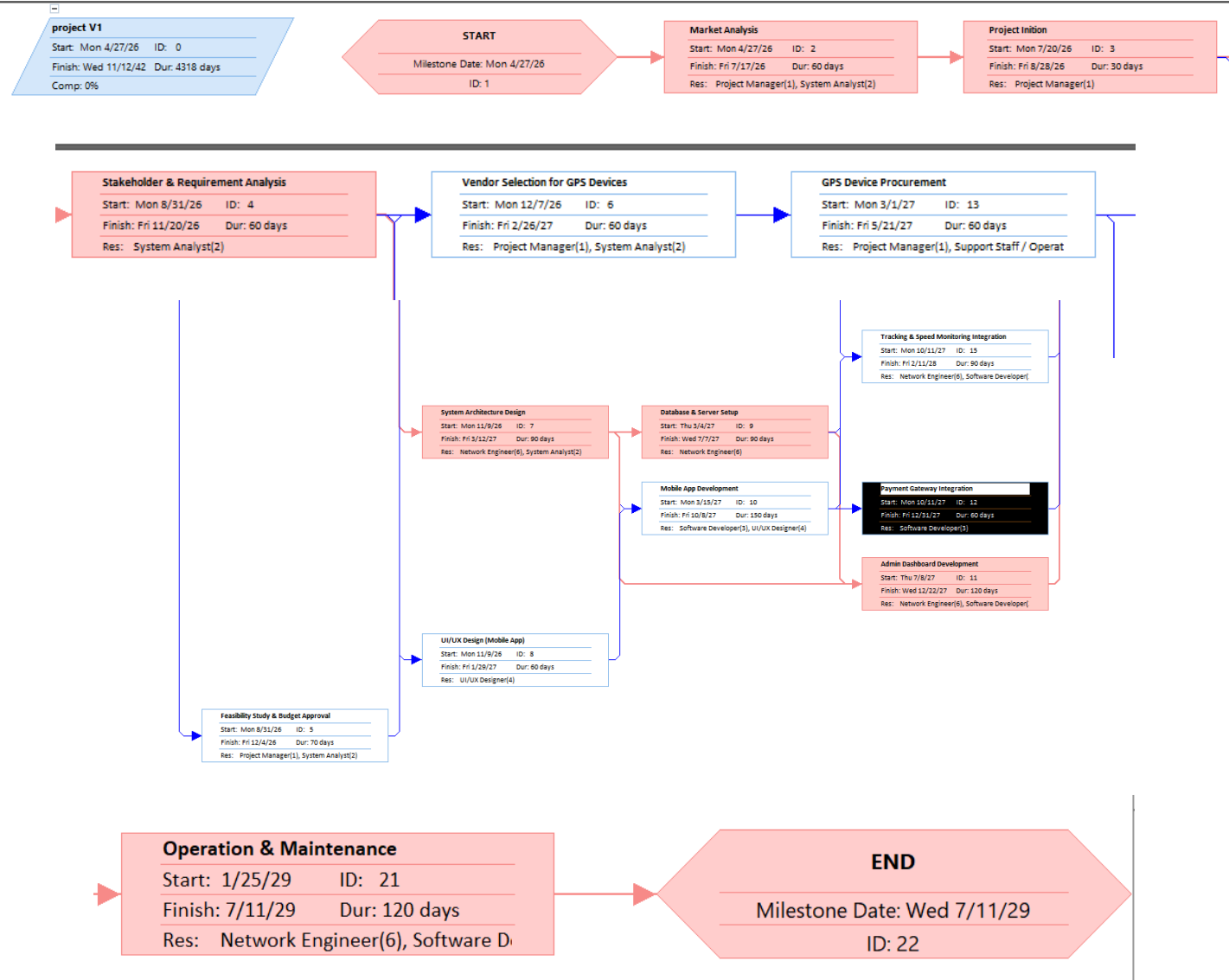


Fig 2 : Network Diagram

- The network diagram clearly illustrates the logical flow of the project from initiation to deployment. It highlights dependencies between tasks and ensures that no activity begins without the completion of prerequisite tasks.
- The diagram also helps identify parallel paths and potential bottlenecks in the workflow. By analyzing the network structure, project managers can focus on optimizing task sequencing and improving overall efficiency.

5.3 Critical Path Identification

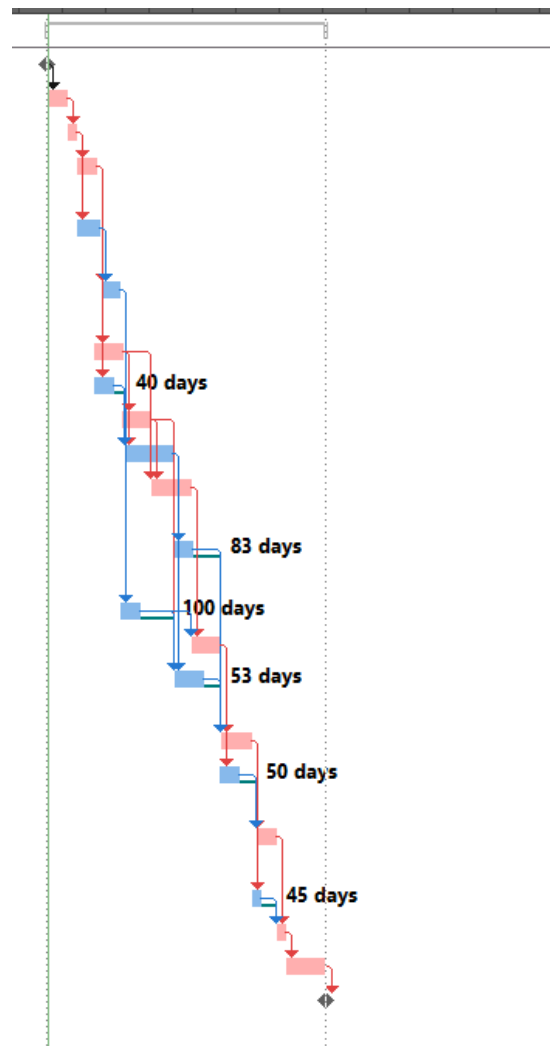


Fig 3 : Critical Path

The identified critical path is:

START → 2 → 3 → 4 → 6 → 7 → 10 → 12 → 16 → 18 → 20 → 21 → END

This path represents the longest sequence of dependent activities and determines the minimum project completion time.

All tasks on this path have **zero slack**, meaning any delay in these activities will directly delay the entire project.

Tasks such as UI/UX Design and Database Setup are not on the critical path, indicating they have scheduling flexibility.

Therefore, special attention must be given to monitoring and managing critical path activities to ensure timely project completion.

5.4 Resource Allocation

5.4.1. Resource sheet :

Resource Name ▼	Type ▼	Material ▼	Initials ▼	Group ▼	Max. ▼	Std. Rate ▼
Project Manager(1)	Work		P		100%	70,000.00₹/mon
System Analyst(2)	Work		S		200%	50,000.00₹/mon
Software Developer(3)	Work		S		400%	50,000.00₹/mon
UI/UX Designer(4)	Work		U		200%	40,000.00₹/mon
Tester(5)	Work		T		300%	30,000.00₹/mon
Network Engineer(6)	Work		N		200%	45,000.00₹/mon
Field Technician (GPS Installation)(7)	Work		F		600%	25,000.00₹/mon
Support Staff / Operator(8)	Work		S		500%	20,000.00₹/mon
Data Entry / Admin(9)	Work		D		200%	22,000.00₹/mon
Trainer(10)	Work		T		100%	35,000.00₹/mon

Fig 4: Resource Sheet

5.4.2 Manpower Allocation :

Sl no	Activity Name	Duration	Manpower (number)
1	START	0 days	-
2	Market Analysis	60 days	1(1), 2(1)
3	Project Initiation	30 days	1(1)
4	Stakeholder & Requirement Analysis	60 days	2(2)
5	Feasibility Study & Budget Approval	60 days	1(1), 2(1)
6	Vendor Selection for GPS Devices	60 days	1(1), 2(1)
7	System Architecture Design	90 days	2(1), 6(1)
8	UI/UX Design (Mobile App)	60 days	4(2)

9	Database & Server Setup	90 days	6(2)
10	Mobile App Development	150 days	3(3), 4(1)
11	Admin Dashboard Development	120 days	3(2), 6(1)
12	Payment Gateway Integration	60 days	3(2)
13	GPS Device Procurement	60 days	1(1), 8(1)
14	GPS Installation (Pilot Buses)	90 days	7(4), 6(1)
15	Tracking & Speed Monitoring Integration	90 days	3(2), 6(1)
16	System Testing	90 days	5(3)
17	User Training	60 days	10(1), 8(2)
18	Pilot Deployment	60 days	3(1), 6(1), 7(2)
19	Documentation	60 days	9(2)
20	Final Review & Closing	30 days	1(1), 2(1)
21	Operation & Maintenance	180 days	3(1), 6(1), 8(2)
22	END	0 days	-

START	0 days			
Market Analysis	60 days	1		Project Manager(1),System Analyst(2)
Project Initiation	30 days	2		Project Manager(1)
Stakeholder & Requirement Analysis	60 days	3		System Analyst(2)
Feasibility Study & Budget Approval	70 days	3		Project Manager(1),System Analyst(2)
Vendor Selection for GPS Devices	60 days	4,5		Project Manager(1),System Analyst(2)
System Architecture Design	90 days	4FS-10 days		Network Engineer(6),System Analyst(2)
UI/UX Design (Mobile App)	60 days	4FS-10 days		UI/UX Designer(4)
Database & Server Setup	90 days	7FS-7 days		Network Engineer(6)
Mobile App Development	150 days	7,8FS-10 day		Software Developer(3),UI/UX Designer(4)
Admin Dashboard Development	120 days	7,9		Network Engineer(6),Software Developer(3)
Payment Gateway Integration	60 days	10		Software Developer(3)
GPS Device Procurement	60 days	6		Project Manager(1),Support Staff / Operator(8)
GPS Installation (Pilot Buses)	90 days	11,13		Field Technician (GPS Installation)(7),Network Engineer(6)
Tracking & Speed Monitoring Integration	90 days	9,10,13		Network Engineer(6),Software Developer(3)
System Testing	90 days	12,14,15		Tester(5)
User Training (Drivers & Admin)	60 days	14FS-5 days		Support Staff / Operator(8),Trainer(10)
Pilot Deployment (50 buses/company)	60 days	16FS+15 days,17		Field Technician (GPS Installation)(7), Network Engineer(6),Software Developer(3)
Documentation	30 days	16		Data Entry / Admin(9)
Final Review and Closing	30 days	18,19		Project Manager(1),System Analyst(2)
Operation & Maintenance	120 days	20		Network Engineer(6),Software Developer(3),Support Staff / Operator(8)
END	0 days	21		

Fig 5: Manpower Allocation

5.5 Resource Overallocation and Leveling

5.5.1 Resource Overallocation

START	0 days		Mon 4/27/26	Mon 4/27/26	
Market Analysis	60 days	1	Mon 4/27/26	Fri 7/17/26	Project Manager(1),System Analyst(2)
Project Initiation	30 days	2	Mon 7/20/26	Fri 8/28/26	Project Manager(1)
Stakeholder & Requirement Analysis	60 days	3	Mon 8/31/26	Fri 11/20/26	System Analyst(2)
Feasibility Study & Budget Approval	60 days	3	Mon 8/31/26	Fri 11/20/26	Project Manager(1), System Analyst(2)
Vendor Selection for GPS Devices	60 days	4,5	Mon 11/23/26	Fri 2/12/27	Project Manager(1), System Analyst(2)
System Architecture Design	90 days	4FS-10 days	Mon 11/9/26	Fri 3/12/27	Network Engineer(6),System Analyst(2)
UI/UX Design (Mobile App)	60 days	4FS-10 days	Mon 11/9/26	Fri 1/29/27	UI/UX Designer(4)
Database & Server Setup	90 days	7FS-7 days	Thu 3/4/27	Wed 7/7/27	Network Engineer(6)
Mobile App Development	150 days	7,8FS-10 days	Mon 3/15/27	Fri 10/8/27	Software Developer(3),UI/UX Designer(4)
Admin Dashboard Development	120 days	7,9	Thu 7/8/27	Wed 12/22/27	Network Engineer(6),Software Developer(3)
Payment Gateway Integration	60 days	10	Mon 10/11/27	Fri 12/31/27	Software Developer(3)
GPS Device Procurement	60 days	6	Mon 2/15/27	Fri 5/7/27	Project Manager(1),Support Staff / Operator(8)
GPS Installation (Pilot Buses)	90 days	11,13	Thu 12/23/27	Wed 4/26/28	Field Technician (GPS Installation)(7),Network Engineer(6)
Tracking & Speed Monitoring Integration	90 days	9,10,13	Mon 10/11/27	Fri 2/11/28	Network Engineer(6), Software Developer(3)
System Testing	90 days	12,14,15	Thu 4/27/28	Wed 8/30/28	Tester(5)
User Training (Drivers & Admin)	60 days	14FS-5 days	Thu 4/20/28	Wed 7/12/28	Support Staff / Operator(8)
Pilot Deployment (50 buses/company)	60 days	16FS+15 days,17	Thu 9/21/28	Wed 12/13/28	Field Technician (GPS Installation)(7), Network Engineer(6),Software Developer(3)

	Resource Name	Type	Material	Initials	Group	Max.	Std. Rate
1	Project Manager(1)	Work		P		100%	70,000.00₹/mon
2	System Analyst(2)	Work		S		200%	50,000.00₹/mon
3	Software Developer(3)	Work		S		400%	50,000.00₹/mon
4	UI/UX Designer(4)	Work		U		200%	40,000.00₹/mon
5	Tester(5)	Work		T		300%	30,000.00₹/mon
6	Network Engineer(6)	Work		N		200%	45,000.00₹/mon
7	Field Technician (GPS Installation)(7)	Work		F		600%	25,000.00₹/mon
8	Support Staff / Operator(8)	Work		S		500%	20,000.00₹/mon
9	Data Entry / Admin(9)	Work		D		2%	22,000.00₹/mon
10	Trainer(10)	Work		T		100%	35,000.00₹/mon

Fig 6: Resource Overallocation

Resource overallocation occurs when a resource is assigned to tasks beyond its available capacity within the same time period.

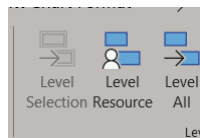
In this project, overallocation was identified specifically in the **System Analyst** resource. This occurred because the System Analyst was assigned to multiple overlapping tasks such as:

- Stakeholder & Requirement Analysis
- Feasibility Study & Budget Approval
- Vendor Selection for GPS Devices

Due to these overlapping assignments, the workload exceeded the resource's maximum capacity, resulting in overallocation.

Microsoft Project highlighted this issue using overallocation indicators, allowing early identification of the scheduling conflict.

5.5.2 Resource Leveling



To resolve the overallocation of the System Analyst, resource leveling was applied using Microsoft Project.

The following actions were taken:

- Task schedules were adjusted to remove overlaps
- Some dependent tasks were delayed slightly
- Workload was redistributed over time

After applying resource leveling:

- The overallocation issue was successfully resolved
- The System Analyst's workload became balanced
- The project schedule was adjusted without affecting critical tasks significantly

Although resource leveling introduced minor delays in some non-critical activities, it ensured a more realistic and manageable project execution plan.

6. Swot Analysis:

Overview:

This SWOT analysis assesses the GPS Bus Tracking System project using MS Project resource sheet, Gantt chart tasks and budget. The SWOT analysis breaks the project into four key elements - Strengths and Weaknesses (internal factors - controllable) and Opportunities and Threats (external factors - uncontrollable).

Project Duration: April 24, 2026 - June 6, 2026 (Development Phase) + Operation & Maintenance

STRENGTHS | Internal - Controllable

1. Structured & Phased Execution

The Gantt chart reveals an orderly project sequence - Initiation & Feasibility Study, Pilot Launch and Operation & Maintenance (O&M) - of well planned project implementation and reduced risk.

2. Specialized Multi-Role Team

The resource sheet verifies that there are six roles in the project:

- Project Manager (70,000₪/month)
- System Analyst (50,000₪/month)
- Software Developer (50,000₪/month)
- UI/UX Designer (40,000₪/month)
- Tester (30,000₪/month)
- Network Engineer (45,000₪/month)

This division of responsibility ensures domain expertise at each phase rather than overburdening generalists.

3. Dual-Interface Design

The project covers both a Mobile App and an Admin Dashboard, covering both end users (public transport riders) and administrators - a good symptom of a complete and deliverable solution.

4. Pilot-First Deployment Strategy

Task 18 notes rollout starts with 50 buses. Slowly increasing the user base lets bug fixing and performance tuning happen before large-scale deployment, thus de-risking large roll-outs.

5. Integrated Payment Gateway

The inclusion of a Payment Gateway (Task 12) indicates revenue potential and possibility for future monetization, enhancing the project's feasibility as a sustainable business.

WEAKNESSES | Internal — Controllable

1. Extremely Tight Development Timeline

The project should begin on April 24, 2026 and end on June 6, 2026 - approximately 6 weeks for development, testing and training. This timeframe is extremely tight, given the project complexity (GPS device, mobile application, web server, administrative dashboard, payment gateway).

2. No Contingency Buffer Visible

All tasks have Variance = 0.00, their Baseline is the same as the Total Cost. There's no margin for contingencies or unexpected costs in the budget - a critical failure in a hardware-containing project.

3. O&M Cost Dominates the Entire Budget

The cost for Task 21 (Operation & Maintenance) is 770,000€ - far exceeding the total cost of all other tasks. The budgeted cost is arguably an annualised recurring cost, but this will be difficult to sustain financially unless there is an assured revenue or subsidy stream.

4. Actual Costs Not Being Tracked

Every column for 'Actual' cost is 0.00€. This either implies that the project hasn't begun, or that costs are not being tracked in real-time (reducing the effectiveness of project control and forecast).

5. Small Pilot Scale

50 buses is a small sample. It might not yield sufficient data to verify system performance with actual city traffic loads and might reveal unexpected problems when fully deployed.

OPPORTUNITIES | External - Respond To

1. Smart City & Digital Transport Initiatives in Bangladesh

Dhaka, and other cities, are building public transport. GPS tracking for bus transportation matches government infrastructure plans and could potentially receive government or city council investment.

2. Scalability of the Platform

The proposed pilot design (an app, server, GPSs and dashboard) is scalable. After initial piloting with 50 buses, city-wide roll-out or licensing to other operators requires little redevelopment effort.

3. Data Monetization & Analytics

Data from speed monitoring and GPS tracking provides significant data. Such data can be used for customer route planning, fleet operation analysis, and even for partnerships with third-party research institutes.

4. Rising Mobile Penetration

Increasing smartphone and mobile data subscriptions in Bangladesh ensure the potential market for a commuter mobile app is likely to be large, and growing.

5. Payment Gateway as a Revenue Channel

The payment gateway enables options like fares collection or user subscription or even advertising revenues, making the system a profitable business venture.

THREATS | External - Beyond Control

1. GPS Hardware Dependency & Vendor Risk

GPS Device Procurement (Task 13) and the resource sheet listing GPS Device at 6,000৳/unit demonstrate outsourcing. Agile delivery risks include supply chain problems, price escalation and device malfunctioning, leading to delivery delays or cost blow-outs.

2. Internet/Data Service Reliability

On the resource sheet, there's an item of 'Internet/Data Service' as a resource and the Database & Server task is by far the second-largest (161,250৳). In areas where Internet service is not always reliable (such as in Dhaka), GPS real-time tracking may face service interruptions.

3. Resistance from Drivers and Operators

Speed monitoring (Task 15) might be viewed as a form of driver surveillance. If not appropriately managed (Task 4 - Stakeholder and Requirement), it is likely that bus operators will resist embracing the change.

4. Cybersecurity & Data Privacy Risks

The system, which deals with data on locations and payments, is a prime target. Any security failure could negatively affect public perception and lead to legal repercussions - particularly as Bangladesh's cybersecurity laws are in transition.

5. Competitive Alternatives

There are also ride-hailing and other third-party transport apps (e.g. Google Maps with transit data) that offer some real-time data for buses. If they provide greater value and are free, then the latter's adoption could be lagging.

Summary Matrix

	Helpful	Harmful
Internal	Clear timeline, roles and responsibilities, pilot project, built-in payment feature	6 weeks to completion, no contingency fund, O&M cost is the bulk of cost, no actual cost tracking
External	Advancement in smart city, scalability of the platform, data analytics opportunities, growth in mobile usage	Vendor hardware risk, network, driver acceptance, cybersecurity risk

Key Strategic Insight

The strategic issue of highest importance relates to the W-T overlap: the pressure of the 6-week delivery time frame, coupled with the risk of hardware and connectivity issues, means the project is likely to slip. Without a contingency budget, a slip could cause a crisis for costs, especially with the 770,000

Recommended Priority Actions:

- Provide longer development time or 10-15% contingency budget
- Engage stakeholders - bus drivers and operators - early to increase cooperation
- Sign a Vendor SLA for the GPS hardware before buying it
- Include real-time cost monitoring in MS Project to avoid cost surprises
- Identify a revenue model to allocate the high O&M cost

Conclusion:

Smart Bus Tracking Speed Monitoring and Digital Ticketing System provides realistic technological solutions to overhaul existing transportation conditions and maintain discipline among passengers, drivers, and bus companies. The proposed system is dedicated to overcome issues like reducing passenger waiting time, knowing realtime bus location, controlling bus speeding, overhand ticketing issues, and drivers behavior while they are on duty.

Passengers will gain realtime bus location, see ETO(bus ETA), monitor bus speeding and able to purchase tickets hassle-free through bKash or Nagad Payment Gateway using GPS tracking system, bus speed monitoring system, digital ticketing system, mobile Application, Admin dashboard, and Digital payment gateway. Bus companies will gain efficient bus operation, transparency of ticketing systems and awareness of driver activities. Traffic authorities like BRTA can support this system by enforcing safer driving and digital payment systems.

After proper investigation and doing project feasibility analysis like project objective, stakeholder, risk analysis, resource planning and scheduling, the project is considered doable and sustainable. As illustrated in stakeholder analysis, passengers, transport companies, authority, drivers, payment gateway providers and IT teams are the key players who want to achieve a goal. Risk analysis also depicts that the risk like GPS sensor malfunction, poor network coverage, fewer users, budget constraint, security threat can occur in the project but it can be overcome by mentioned actions.

To conclude, the SMART BUS project is feasible to implement and useful to our digital Bangladesh vision. If the pilot version goes successfully by implementing 50 buses per transport company then public inconvenience will reduce, road safety will increase, Digital payment will be popularized and digital platforms will be made for Bangladesh Transport. With time, we can add more routes to this project and include more transport companies. The ultimate goal will be to apply this system to every part of Bangladesh.

Member is responsible for:

Name	Responsibility
Md. Shahmul Islam (2022-3-60-154)	1) Develop project Charter 2) Perform Stakeholder Analysis
Rukaiah Bintay Morshed (2022-3-60-334)	3) Perform Risk Analysis
Farha Anjum Afifa (2022-3-60-237)	4) Cost Benefit Analysis
Sadman Tayef (2022-3-50-002)	5)Project Gantt Chart preparation, Critical Path Identification, Network Diagram, Resource Allocation
Hasnay Hasin (2022-3-60-038)	6) SWOT analysis of the project